

MADEIRA AP, Portugal

WMO: 085210

Lat: 32.693N

Lon: 16.776W

Elev: 58

StdP: 100.63

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.3	13.0	3.9	5.0	17.5	5.3	5.5	15.7	15.0	15.5	13.4	15.9	3.9	350	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	27.7	19.9	26.4	20.2	25.6	20.1	22.2	25.4	21.5	24.8	21.0	24.4	5.9	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	15.8	24.4	20.2	15.0	23.9	19.6	14.4	23.6	65.1	25.5	62.7	24.8	60.8	24.4	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	11.6	10.2	10.7	30.8	1.7	2.5	9.5	32.6	8.5	34.1	7.5	35.5	6.3	37.3
			7.6	23.0	1.2	0.6	6.7	23.4	6.0	23.8	5.3	24.1	4.5	24.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.5	16.5	16.2	16.7	17.4	18.7	20.6	22.4	23.4	22.8	21.6	19.3	17.8
	DBStd	2.88	1.43	1.37	1.57	1.42	1.48	1.38	1.45	1.20	1.21	1.39	1.65	1.36
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	258	58	61	55	35	13	1	0	0	0	1	9	27
	CDD10.0	3457	203	174	208	222	269	319	384	414	386	359	279	241
	CDD18.3	673	2	2	4	7	23	69	126	156	136	101	38	9
	CDH23.3	1373	0	0	3	5	16	75	307	501	338	119	10	0
(14) Wind	WSAvg	4.8	4.6	4.9	5.4	4.9	4.6	4.8	5.4	5.1	4.3	4.4	4.5	4.6
	PrecAvg	1523	188	180	153	114	71	37	10	17	88	191	212	262
	PrecMax	2272	394	583	317	330	249	138	20	34	288	293	392	516
	PrecMin	813	54	7	57	32	16	5	1	6	26	58	48	79
	PrecStd	251	89	103	59	60	46	24	5	6	49	65	74	112
	DB	21.2	21.6	23.1	23.8	25.1	27.0	30.9	30.2	29.3	27.0	24.3	22.2	
	MCWB	16.9	15.2	14.7	15.8	16.6	18.5	17.9	18.4	18.8	20.4	19.4	17.1	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	20.2	20.2	21.2	21.7	23.2	25.0	27.3	27.3	26.7	25.3	23.4	21.2	
	2% MCWB	16.5	15.7	15.6	15.5	17.7	19.1	19.0	20.5	20.8	20.0	19.1	17.1	
	5% DB	19.4	19.2	20.1	20.7	22.2	24.0	25.7	26.3	25.8	24.5	22.6	20.6	
	5% MCWB	15.9	15.2	15.6	15.6	17.4	19.0	19.7	20.6	20.6	19.6	18.7	16.9	
	10% DB	18.7	18.5	19.2	19.9	21.2	23.2	24.8	25.6	25.1	23.8	21.9	20.0	
	10% MCWB	15.4	14.7	15.1	15.4	16.9	18.5	19.4	20.3	20.1	19.4	18.2	16.6	
	WB	18.3	18.0	18.2	18.2	19.6	21.4	21.7	22.9	23.1	22.5	21.1	19.4	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	20.0	19.4	20.5	20.9	22.8	24.5	25.6	26.8	26.2	24.8	22.8	20.6	
	2% WB	17.3	16.9	17.2	17.3	18.7	20.4	21.0	22.0	22.2	21.5	20.3	18.4	
	2% MCDB	19.4	19.2	19.6	20.1	21.8	23.8	24.9	25.8	25.4	23.9	22.5	20.3	
	5% WB	16.5	16.0	16.4	16.7	18.1	19.7	20.5	21.4	21.4	20.8	19.5	17.7	
	5% MCDB	18.9	18.5	19.1	19.6	21.3	23.1	24.4	25.2	24.7	23.4	22.0	19.9	
	10% WB	15.8	15.2	15.6	16.0	17.4	19.0	19.9	20.8	20.7	20.2	18.7	17.0	
	10% MCDB	18.4	17.9	18.5	19.2	20.7	22.5	24.0	24.8	24.2	23.1	21.4	19.5	
(35) Mean Daily Temperature Range	MDBR	3.6	3.7	4.1	4.3	4.4	4.4	4.4	4.4	4.2	4.1	3.7	3.6	
	5% DB	4.4	5.0	5.3	5.3	5.4	5.5	5.6	5.3	5.2	4.9	4.6	4.4	
	5% WB	3.4	3.8	3.9	3.7	3.5	3.5	3.5	3.4	3.4	3.4	3.4	3.5	
	MCDBR	4.1	4.4	4.8	4.8	5.1	5.1	4.9	4.9	4.7	4.2	4.1	3.8	
	MCWBR	3.7	3.8	4.0	3.8	3.8	3.7	3.5	3.5	3.6	3.5	3.6	3.6	
(40) Clear-Sky Solar Irradiance	taub	0.341	0.362	0.390	0.380	0.383	0.382	0.391	0.395	0.391	0.376	0.367	0.348	
	taud	2.549	2.470	2.383	2.432	2.446	2.472	2.449	2.460	2.487	2.523	2.506	2.542	
	Ebn at Noon	878	890	888	910	904	900	890	885	876	866	840	849	
	Edh at Noon	83	99	116	115	114	111	113	110	103	93	86	80	
(44) All-Sky Solar Radiation	RadAvg	2.83	3.65	4.89	6.03	6.58	7.04	7.01	6.56	5.52	4.21	3.01	2.54	
	RadStd	0.14	0.25	0.25	0.22	0.42	0.35	0.30	0.24	0.18	0.18	0.18	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+0.60	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page